Backend Overview

The backend is a Node.js + Express 5 REST API that powers the KUCCPS IT Ticketing System.

Quick Facts

Aspect	Detail
Runtime	Node.js 20+
Framework	Express 5
ORM	Prisma 5
Database	PostgreSQL 14+
Auth	JWT + bcrypt
Port	5000 (default)

What the Backend Does

- Serves the REST API for all frontend operations
- Manages database access through Prisma ORM
- Handles authentication and authorization
- Sends email notifications via SMTP
- Monitors inbound email via IMAP
- Syncs users from Active Directory
- Runs scheduled background jobs

Key Entry Points

File	Purpose
`src/server.js`	Application entry point, middleware, route mounting
`src/routes/*.js`	API route definitions
`src/controllers/*.js`	Request handlers
`src/services/*.js`	Business logic and integrations
`src/middleware/*.js`	Auth, security, and validation

Base URL

```
http://localhost:5000/api
```

Documentation Sections

- Server Setup — Initialization and middleware
- Middleware — Auth, security, and file upload
- Controllers — Request handlers
- Routes — API endpoint definitions
- Services — Email, AD, and scheduler
- Security — Authentication, authorization, and hardening
- Scheduler — Background jobs

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